

Calculus and Linear Algebra

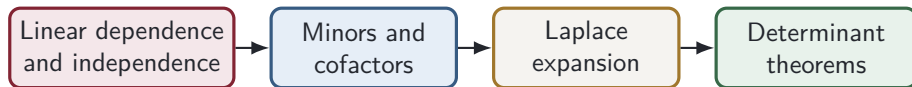
Linear Dependence and Determinants

Lecture 5

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Lecture outline



Aim

To become completely comfortable with linear dependence and independence, and then connect these ideas with determinants.

Linear dependence: what are we checking?

Let

$$v_1, \dots, v_k \in \mathbb{R}^n.$$

The vectors are **linearly dependent** if there exist scalars

$$\lambda_1, \dots, \lambda_k,$$

not all equal to zero, such that

$$\lambda_1 v_1 + \dots + \lambda_k v_k = 0.$$

They are **linearly independent** if the only possibility is

$$\lambda_1 = \dots = \lambda_k = 0.$$

The key question is always: can a non-trivial linear combination give the zero vector?

A practical strategy

Suppose we are given three vectors in $\mathbb{R}^3$:

$$v_1 = (a_1, a_2, a_3), \quad v_2 = (b_1, b_2, b_3), \quad v_3 = (c_1, c_2, c_3).$$

Write

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

Comparing coordinates gives a system:

$$\begin{cases} \alpha a_1 + \beta b_1 + \gamma c_1 = 0, \\ \alpha a_2 + \beta b_2 + \gamma c_2 = 0, \\ \alpha a_3 + \beta b_3 + \gamma c_3 = 0. \end{cases}$$

Only the zero solution $\Rightarrow$ independent. A non-zero solution $\Rightarrow$ dependent.

Exercise: immediate dependence in $\mathbb{R}^3$

Exercise

Consider

$$v_1 = (1, 2, -1), \quad v_2 = (2, 4, -2), \quad v_3 = (0, 1, 3).$$

Are v_1, v_2, v_3 linearly independent?

Before writing any system, **inspect the vectors carefully.**

Exercise: solution

Solution

We immediately see that

$$v_2 = 2v_1.$$

Therefore

$$2v_1 - v_2 + 0v_3 = 0,$$

and the coefficients are not all zero.

Hence

v_1, v_2, v_3 are linearly dependent.

A single proportional pair is enough to make the whole family dependent.

Exercise: one vector from the others

Exercise

Let

$$v_1 = (1, 0, 2), \quad v_2 = (0, 1, -1), \quad v_3 = (2, 1, 3).$$

Determine whether the three vectors are linearly dependent or independent.

Can you express one vector as a linear combination of the other two?

Exercise: solution

Solution

Observe that

$$2v_1 + v_2 = 2(1, 0, 2) + (0, 1, -1) = (2, 1, 3) = v_3.$$

Thus

$$2v_1 + v_2 - v_3 = 0.$$

This is a non-trivial linear relation, so

v_1, v_2, v_3 are linearly dependent.

Exercise: independence in $\mathbb{R}^3$

Exercise

Consider

$$v_1 = (1, 0, 1), \quad v_2 = (0, 1, 1), \quad v_3 = (1, 1, 0).$$

Study their linear independence by starting from

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

Write the coordinate equations explicitly.

Exercise: solution

Solution

From

$$\alpha(1, 0, 1) + \beta(0, 1, 1) + \gamma(1, 1, 0) = (0, 0, 0),$$

we obtain

$$\begin{cases} \alpha + \gamma = 0, \\ \beta + \gamma = 0, \\ \alpha + \beta = 0. \end{cases}$$

The first two equations give $\alpha = \beta = -\gamma$. Substituting into the third,

$$-2\gamma = 0 \quad \Rightarrow \quad \gamma = 0.$$

Hence $\alpha = \beta = \gamma = 0$, and therefore

v_1, v_2, v_3 are linearly independent.

Exercise: find the dependence relation

Exercise

Let

$$v_1 = (1, 2, 0), \quad v_2 = (0, 1, 1), \quad v_3 = (2, 5, 1).$$

The vectors are linearly dependent.

Find explicitly a non-trivial relation

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

Exercise: solution

Solution

We write

$$\alpha(1, 2, 0) + \beta(0, 1, 1) + \gamma(2, 5, 1) = 0,$$

so

$$\begin{cases} \alpha + 2\gamma = 0, \\ 2\alpha + \beta + 5\gamma = 0, \\ \beta + \gamma = 0. \end{cases}$$

Take $\gamma = 1$. Then

$$\alpha = -2, \quad \beta = -1.$$

Thus

$$\boxed{-2v_1 - v_2 + v_3 = 0.}$$

Equivalently, $v_3 = 2v_1 + v_2$.

Exercise: four vectors in $\mathbb{R}^4$

Exercise

Consider

$$\begin{aligned}v_1 &= (1, 0, 0, 1), & v_2 &= (0, 1, 0, 1), \\v_3 &= (0, 0, 1, 1), & v_4 &= (1, 1, 1, 3).\end{aligned}$$

Are these four vectors linearly independent?

Try inspection first. Then write a linear relation if one exists.

Exercise: solution

Solution

Adding the first three vectors gives

$$v_1 + v_2 + v_3 = (1, 1, 1, 3) = v_4.$$

Therefore

$$v_1 + v_2 + v_3 - v_4 = 0.$$

Since the coefficients are not all zero,

v_1, v_2, v_3, v_4 are linearly dependent.

Exercise: independence in $\mathbb{R}^4$

Exercise

Let

$$\begin{aligned}v_1 &= (1, 0, 0, 1), & v_2 &= (0, 1, 0, 1), \\v_3 &= (0, 0, 1, 1), & v_4 &= (0, 0, 0, 1).\end{aligned}$$

Starting from

$$\alpha v_1 + \beta v_2 + \gamma v_3 + \delta v_4 = 0,$$

determine whether the four vectors are linearly independent.

Exercise: solution

Solution

Comparing coordinates gives

$$\begin{cases} \alpha = 0, \\ \beta = 0, \\ \gamma = 0, \\ \alpha + \beta + \gamma + \delta = 0. \end{cases}$$

The first three equations imply

$$\alpha = \beta = \gamma = 0,$$

and then the last equation gives $\delta = 0$.

Hence the only solution is the trivial one:

v_1, v_2, v_3, v_4 are linearly independent.

Exercise: a less immediate case in $\mathbb{R}^4$

Exercise

Consider

$$v_1 = (1, 1, 0, 0), \quad v_2 = (0, 1, 1, 0),$$

$$v_3 = (0, 0, 1, 1), \quad v_4 = (1, 2, 2, 1).$$

Determine whether the four vectors are linearly dependent.

If they are, find an explicit dependence relation.

Exercise: solution

Solution

We look for

$$\alpha v_1 + \beta v_2 + \gamma v_3 + \delta v_4 = 0.$$

The coordinate equations are

$$\begin{cases} \alpha + \delta = 0, \\ \alpha + \beta + 2\delta = 0, \\ \beta + \gamma + 2\delta = 0, \\ \gamma + \delta = 0. \end{cases}$$

Taking $\delta = 1$ gives

$$\alpha = -1, \quad \beta = -1, \quad \gamma = -1.$$

Therefore

$$\boxed{-v_1 - v_2 - v_3 + v_4 = 0,}$$

Exercise: dependence with a parameter

Exercise

Let

$$v_1 = (1, 0, 1), \quad v_2 = (0, 1, 1), \quad v_3 = (1, 1, a).$$

For which value of a are the vectors linearly dependent?

Start from

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

Exercise: solution

Solution The coordinate equations are

$$\begin{cases} \alpha + \gamma = 0, \\ \beta + \gamma = 0, \\ \alpha + \beta + a\gamma = 0. \end{cases}$$

From the first two equations,

$$\alpha = -\gamma, \quad \beta = -\gamma.$$

Hence the third becomes

$$(a - 2)\gamma = 0.$$

A non-trivial solution exists exactly when

$$a = 2.$$

Indeed, for $a = 2$, $v_3 = v_1 + v_2$.

Quick practice: dependent or independent?

Exercise

Decide as quickly as possible.

1. $(1, 2, 3), (2, 4, 6)$ in $\mathbb{R}^3$.
2. $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ in $\mathbb{R}^3$.
3. $(1, 0, 1, 0), (0, 1, 0, 1), (1, 1, 1, 1)$ in $\mathbb{R}^4$.
4. $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)$ in $\mathbb{R}^4$.

For each case, state the reason, not only the answer.

Quick practice: solution

Solution

1. Dependent, because the second vector is twice the first.
2. Independent: each coordinate direction is represented separately.
3. Dependent, because

$$(1, 1, 1, 1) = (1, 0, 1, 0) + (0, 1, 0, 1).$$

4. Independent: the only combination of the four standard basis vectors giving 0 has all coefficients equal to zero.

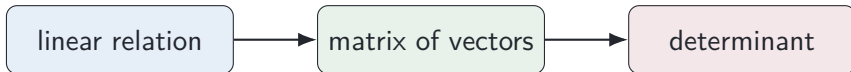
Look for an obvious relation first. Write the system only when inspection is not enough.

From linear dependence to determinants

The previous exercises were solved directly from

$$\lambda_1 v_1 + \cdots + \lambda_k v_k = 0.$$

For square collections of vectors, the determinant will provide a compact test for the same question.



Recall: the 2×2 determinant

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det A = ad - bc.$$

If the columns of A are vectors in $\mathbb{R}^2$, then

$$|\det A| = \text{area of the generated parallelogram.}$$

Moreover,

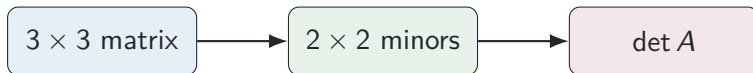
$$\det A = 0 \iff \text{the two columns are linearly dependent.}$$

From 2×2 to larger matrices

For a 3×3 matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

we reduce the computation to determinants of 2×2 submatrices.



Minor associated with an entry

Definition

For a square matrix $A = (a_{ij})$, the minor M_{ij} is the determinant obtained by deleting row i and column j .

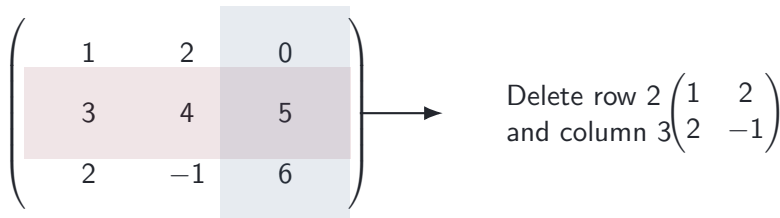
Example:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 2 & -1 & 6 \end{pmatrix}.$$

Deleting row 2 and column 3 gives

$$M_{23} = \det \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = -5.$$

A minor: row and column deletion



The diagram illustrates the process of finding a minor by deleting a row and a column from a matrix. On the left, a 3x3 matrix is shown with its elements: $\begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 2 & -1 & 6 \end{pmatrix}$. The second row (3, 4, 5) is highlighted in light red, and the third column (0, 5, 6) is highlighted in light blue. An arrow points from the matrix to the right, where the text "Delete row 2 and column 3" is written, followed by the resulting 2x2 matrix: $\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$.

Cofactors

Definition

The cofactor associated with a_{ij} is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The signs follow the pattern

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Hence, for the previous example,

$$C_{23} = (-1)^5 M_{23} = 5.$$

Exercise: minors and cofactors

Exercise

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix}.$$

Compute

$$M_{11}, \quad M_{12}, \quad M_{23},$$

and the corresponding cofactors

$$C_{11}, \quad C_{12}, \quad C_{23}.$$

Exercise: solution

Solution

$$M_{11} = \det \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} = -9, \quad C_{11} = -9.$$

$$M_{12} = \det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} = -20, \quad C_{12} = 20.$$

$$M_{23} = \det \begin{pmatrix} 2 & 1 \\ 5 & 2 \end{pmatrix} = -1, \quad C_{23} = 1.$$

Laplace expansion along a row

Theorem

For any row i of an $n \times n$ matrix,

$$\det A = \sum_{j=1}^n a_{ij} C_{ij}.$$

For a 3×3 matrix, expanding along the first row gives

$$\det A = a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}.$$

Equivalently,

$$\det A = a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}.$$

Laplace expansion: a complete example

Let

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & -1 & 4 \\ 1 & 2 & 5 \end{pmatrix}.$$

Expanding along the first row,

$$\det A = 2 \det \begin{pmatrix} -1 & 4 \\ 2 & 5 \end{pmatrix} - 1 \det \begin{pmatrix} 3 & 4 \\ 1 & 5 \end{pmatrix} + 0 \cdot M_{13}.$$

Therefore

$$\det A = 2(-13) - 11 = -37.$$

Laplace expansion along a column

The expansion can be performed along any column as well:

$$\det A = \sum_{i=1}^n a_{ij} C_{ij}.$$

For

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & -1 & 4 \\ 1 & 2 & 5 \end{pmatrix},$$

expanding along the third column is possible, but the first row is shorter because it contains a zero.

Choose a row or column containing as many zeros as possible.

Laplace expansion: choose the convenient line

Consider

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ -1 & 5 & 2 \end{pmatrix}.$$

Expanding along the first row,

$$\det B = 1 \det \begin{pmatrix} 3 & 4 \\ 5 & 2 \end{pmatrix} = 6 - 20 = -14.$$

No other minors need to be computed.

Exercise: determinant by Laplace expansion

Exercise

Compute the determinant of

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 0 & 5 & 2 \end{pmatrix}$$

using Laplace expansion.

Choose the row or column that makes the computation shortest.

Exercise: solution

Solution Expand along the first column:

$$\det A = 1 \det \begin{pmatrix} -1 & 4 \\ 5 & 2 \end{pmatrix} + 3C_{21}.$$

Since

$$C_{21} = -\det \begin{pmatrix} 2 & 0 \\ 5 & 2 \end{pmatrix} = -4,$$

we obtain

$$\det A = (-22) + 3(-4) = \boxed{-34}.$$

A 4×4 determinant

Laplace expansion is recursive: an $n \times n$ determinant is reduced to determinants of order $n - 1$.

For example,

$$A = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 4 & 1 & 2 & 0 \\ 0 & 0 & 5 & 1 \end{pmatrix}.$$

The second row contains three zeros, so expanding along it reduces the problem immediately to one 3×3 determinant.

Exercise: a sparse 4×4 matrix

Exercise

Compute

$$\det \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 4 & 1 & 2 & 0 \\ 0 & 0 & 5 & 1 \end{pmatrix}$$

by Laplace expansion.

Indicate clearly the row or column used at each step.

Exercise: solution

Solution Expanding along row 2,

$$\det A = 3 \det \begin{pmatrix} 1 & 0 & 2 \\ 4 & 2 & 0 \\ 0 & 5 & 1 \end{pmatrix}.$$

Expanding the 3×3 determinant along the first row,

$$\det A = 3 \left[1 \det \begin{pmatrix} 2 & 0 \\ 5 & 1 \end{pmatrix} + 2 \det \begin{pmatrix} 4 & 2 \\ 0 & 5 \end{pmatrix} \right] = 3(2 + 40) = \boxed{126}.$$

Triangular matrices

Theorem

If A is upper triangular or lower triangular, then

$$\det A = \prod_{i=1}^n a_{ii}.$$

Example:

$$A = \begin{pmatrix} 2 & 1 & 4 \\ 0 & -3 & 5 \\ 0 & 0 & 6 \end{pmatrix} \implies \det A = 2(-3)6 = -36.$$

Immediate cases

For every n ,

$$\det I_n = 1.$$

For the zero matrix,

$$\det 0 = 0.$$

More generally, any triangular matrix with at least one zero on the main diagonal has determinant zero. **Exercise** Compute mentally:

$$\det \begin{pmatrix} 4 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -2 \end{pmatrix}, \quad \det \begin{pmatrix} 1 & 0 & 0 \\ 5 & 0 & 0 \\ 2 & 4 & 3 \end{pmatrix}.$$

Effect of exchanging two rows

Theorem

If B is obtained from A by interchanging two rows, then

$$\det B = -\det A.$$

The same statement holds for columns.

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \det A = -2,$$

while

$$B = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}, \quad \det B = 2.$$

Multiplying one row by a scalar

Theorem

If one row of A is multiplied by c , then the determinant is multiplied by c .

$$R_i \mapsto cR_i \implies \det B = c \det A.$$

Consequently, multiplying the whole $n \times n$ matrix by c gives

$$\det(cA) = c^n \det A.$$

Adding a multiple of another row

Theorem

The operation

$$R_i \mapsto R_i + cR_j, \quad i \neq j,$$

does not change the determinant:

$$\boxed{\det B = \det A.}$$

This property will be important when row operations are introduced for linear systems.

Determinant of a product

Theorem

For square matrices A and B of the same order,

$$\det(AB) = \det A \det B.$$

In general,

$$\det(A + B) \neq \det A + \det B.$$

The determinant is multiplicative, not additive.

Transpose and determinant

Theorem

For every square matrix A ,

$$\det(A^T) = \det A.$$

Thus every theorem stated in terms of rows has an analogous formulation in terms of columns. In particular,

$$\text{dependent rows} \iff \text{dependent columns}$$

for a square matrix with determinant zero.

Determinant properties: summary

Operation / relation	Effect on determinant
swap two rows	changes sign
$R_i \mapsto cR_i$	multiplied by c
$R_i \mapsto R_i + cR_j$	unchanged
transpose	unchanged
product AB	$\det A \det B$
triangular matrix	product of diagonal entries

Exercise: use properties, not expansion

Exercise

Suppose A is a 3×3 matrix with

$$\det A = -4.$$

Find:

1. $\det(A^\top)$;
2. $\det(2A)$;
3. $\det(A^2)$;
4. the determinant after swapping two rows of A ;
5. the determinant after replacing R_3 by $R_3 + 5R_1$.

Exercise: solution

Solution Since A is 3×3 and $\det A = -4$,

$$\det(A^T) = -4,$$

$$\det(2A) = 2^3(-4) = -32,$$

$$\det(A^2) = (-4)^2 = 16.$$

After one row swap, the determinant is 4.

After $R_3 \mapsto R_3 + 5R_1$, the determinant remains -4 .

Determinant and linear dependence

Theorem

Let A be an $n \times n$ matrix. Then

$$\det A = 0 \iff \text{the columns of } A \text{ are linearly dependent.}$$

Equivalently,

$$\det A = 0 \iff \text{the rows of } A \text{ are linearly dependent.}$$

A structural zero determinant

Consider

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 0 & 6 \\ 2 & 4 & 8 \end{pmatrix}.$$

The third row is twice the first row:

$$R_3 = 2R_1.$$

Hence the rows are linearly dependent, and therefore

$$\boxed{\det A = 0}$$

without any Laplace expansion.

Exercise: detect zero determinants

Exercise

Determine which matrices have determinant zero without computing all determinants explicitly.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 1 \\ 4 & 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 6 & 1 \\ -1 & -2 & 4 \end{pmatrix}.$$

State the reason in each case.

Exercise: solution

Solution For A ,

$$R_2 = 2R_1,$$

so $\det A = 0$.

For C , the first two columns are dependent:

$$C_2 = 2C_1,$$

so $\det C = 0$.

Matrix B has no immediate dependence relation; direct computation gives

$$\det B = -25 \neq 0.$$

Lecture 5: main results

Linear dependence

- ▶ non-trivial linear relations;
- ▶ direct inspection when possible;
- ▶ coordinate systems in $\mathbb{R}^3$ and $\mathbb{R}^4$;
- ▶ parameter-dependent relations.

Determinants

- ▶ minors and cofactors;
- ▶ Laplace expansion;
- ▶ triangular determinants;
- ▶ determinant properties;
- ▶ determinant and dependence.

$\det A = 0 \iff$ rows/columns of a square matrix are linearly dependent.

Preparation for Lecture 6

The next step is to use matrices to represent systems of linear equations.

$$A\mathbf{x} = \mathbf{b}.$$

The work done today on linear combinations and determinants will be the natural starting point.

